
NIFTY 100 FINANCIAL INTELLIGENCE PLATFORM

ANALYST REFERENCE GUIDE

A Comprehensive Manual on KMeans Clustering, Multi-Factor Analytics, and System Integration

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Executive Summary

The Nifty 100 Financial Intelligence Platform represents a state-of-the-art implementation of automated financial analysis. Historically, investment research has been constrained by fragmented data formats, inconsistent fiscal periods, and manual spreadsheet-based computations. This platform solves these pain points by implementing a robust ETL pipeline, standardising financial datasets for 92 conformed companies, running unsupervised KMeans clustering, exposing a high-performance REST API, and providing an interactive Streamlit dashboard.

By automating data extraction, quality validation, KPI computation, and profiling, the platform delivers institutional-grade investment research insights in seconds. The following pages detail the underlying technical framework, mathematical formulas, API specifications, and operational manuals.

1. Research Methodology

The objective of our clustering engine is to segment the Nifty 100 universe into homogeneous financial archetypes based on operational efficiency, growth, leverage, and cash flow generation. The engine utilizes a 5-factor model to capture multi-dimensional corporate profiles before applying unsupervised machine learning.

Clustering Feature Selection

The machine learning model utilizes five core features representing critical pillars of corporate finance:

- **Return on Equity (ROE %)**: Measures capital efficiency and profitability relative to equity capital.
- **Debt-to-Equity (D/E Ratio)**: Captures balance sheet leverage, capital structure risk, and gearing.
- **Revenue CAGR 5yr (%)**: Measures long-term compound growth of top-line sales.
- **Free Cash Flow (FCF) CAGR 5yr (%)**: Measures capital efficiency and growth in distributable cash flow.
- **Operating Profit Margin (OPM %)**: Represents pricing power, cost control, and operational efficiency.

Preprocessing & Standardization

To ensure robust clustering results, raw data must undergo cleaning and preprocessing. First, missing metric values are imputed using the **median value of their broad industry sector** (e.g., IT, Financials, Energy) to preserve industry-specific capital structures. Global medians are applied as a fallback. Second, because KMeans uses Euclidean distance to group records, variables with large absolute scales (like ROE vs. D/E) would dominate the clustering. To prevent this, we apply **StandardScaler**, which normalizes each feature to have a mean of 0 and a standard deviation of 1:

$$z = (x - \mu) / \sigma$$

KMeans is then executed with **n_clusters=5** and **random_state=42** for mathematical reproducibility.

2. Cluster Archetype Profiles

By applying KMeans clustering to the conformed 92-company dataset, five distinct financial archetypes emerged from their multi-factor signatures. This clustering removes subjective bias and groups companies based on actual balance sheet and profit-and-loss characteristics.

High-Quality Compounders (Cluster 0)

Profile: High return on equity (median ~18.5%), stable margins (median ~20.5%), low debt (median D/E ~0.11), and consistent growth.

Sectors: Technology, Consumer Staples, Automobiles, Pharmaceuticals.

Representative Tickers: TCS, INFY, ASIANPAINT, SUNPHARMA, TATAMOTORS.

Defensive Dividend Payers (Cluster 1)

Profile: Extremely high operating profit margins (median ~85.1%), stable cash flows, moderate growth, and high dividend payouts.

Sectors: Utilities, Telecom, Financial Holding Companies, Insurance.

Representative Tickers: COALINDIA, POWERGRID, HDFCLIFE, BHARTIARTL, JIOFIN.

Value Cyclical (Cluster 2)

Profile: Highly leveraged balance sheets (median D/E ~6.9), cyclical earnings, solid asset bases, and capital-intensive structures.

Sectors: Commercial Banks, NBFCs, Heavy Infrastructure, Commodity Producers.

Representative Tickers: HDFCBANK, ICICIBANK, AXISBANK, PFC, IRFC.

Emerging Growth (Cluster 4)

Profile: High revenue compound growth (median ~18.0%), moderate leverage (median ~0.80), active reinvestment, and lower near-term FCF margins.

Sectors: Consumer Discretionary, Retail, Hospital Chains, Specialised Services.

Representative Tickers: ADANIENT, DMART, APOLLOHOSP, TVSMOTOR, TRENT.

Distressed or Turnaround (Cluster 3)

Profile: High free cash flow growth outliers (due to low base effects), lower operational margins, and undergoing restructuring.

Sectors: Energy transmission, public sector infrastructure.

Representative Tickers: GAIL, NHPC.

3. Statistical & Peer Analysis

A critical function of the platform is performing cross-sectional statistical analysis. This includes identifying outliers (anomaly detection) and computing percentile rankings within industry groups to provide context to absolute valuations.

Z-Score Outlier Detection

To flag anomalous data entries or unique corporate structures, we perform **industry-stratified Z-score profiling**. For each company, we calculate the Z-score of its KPIs against the mean (μ) and standard deviation (σ) of its specific sector. A company is flagged as an outlier if its absolute Z-score exceeds 3.0 on any metric:

$$Z = (x - \mu_{\text{sector}}) / \sigma_{\text{sector}}$$

This ensures that utility firms with high debt are not compared against debt-free software companies, reducing false positives. Outliers are saved to `output/outlier_report.csv` for human review.

Peer Group Percentiles

Evaluating a company's performance in isolation is misleading. The platform categorises all 92 companies into 11 distinct peer groups (e.g. IT Services, Private Banks, Oil & Gas). Within each peer group, we compute the **percentile rank** of each company for 10 core metrics. The percentile rank is calculated using the conformed SQL formula:

$$\text{Percentile Rank} = (Rank - 1) / (N - 1)$$

where *Rank* is the 1-based ascending rank of the company's value within the peer group, and *N* is the count of valid companies. This yields a value between 0.0 (weakest) and 1.0 (strongest), allowing analysts to instantly evaluate peer standing.

4. Financial KPI Definitions (Part I)

The platform implements conformed financial definitions to ensure absolute consistency across all reporting layers. The formulas are coded in Python and SQL as follows:

Return on Equity (ROE)

ROE measures a firm's efficiency at generating profits from every rupee of shareholders' equity.

Formula: $ROE = (\text{Net Profit} / \text{Total Equity}) * 100$

where Total Equity = Equity Capital + Reserves & Surplus. If Total Equity is negative, ROE is returned as Null.

Return on Capital Employed (ROCE)

ROCE evaluated the profitability of a company relative to the total capital it employs (debt + equity).

Formula: $ROCE = (\text{EBIT} / \text{Capital Employed}) * 100$

where EBIT = Profit Before Tax + Interest, and Capital Employed = Equity Capital + Reserves + Borrowings.

Net Profit Margin (NPM)

NPM represents the percentage of revenue remaining after all operating, interest, and tax expenses.

Formula: $NPM = (\text{Net Profit} / \text{Sales}) * 100$

Debt-to-Equity (D/E Ratio)

D/E captures capital structure risk. High ratios indicate heavy reliance on debt financing.

Formula: $D/E = \text{Borrowings} / (\text{Equity Capital} + \text{Reserves})$

If the company is debt-free (Borrowings = 0 or Null), D/E is set to 0.0.

Free Cash Flow (FCF)

FCF is the cash generated by operations minus cash spent on capital expenditures (Capex).

Formula: $FCF = \text{CFO} - \text{Capex}$

where CFO is cash from operating activities, and Capex is investing cash outflow on fixed assets.

5. Financial KPI Definitions (Part II)

The platform also calculates growth rates, valuation multiples, and coverage safety factors:

Compound Annual Growth Rate (CAGR)

CAGR represents the annualized growth rate of a metric (Revenue, PAT, or FCF) over a 5-year period.

Formula: $CAGR = (Ending\ Value / Beginning\ Value)^{(1/5)} - 1$

Edge-Case Flags:

- **TURNAROUND:** Flagged when Beginning Value < 0 and Ending Value > 0. CAGR returned as Null.
- **DECLINE_TO_LOSS:** Flagged when Beginning Value > 0 and Ending Value < 0. CAGR returned as Null.
- **BOTH_NEGATIVE:** Flagged when both values are negative. CAGR returned as Null.
- **ZERO_BASE:** Flagged when Beginning Value is 0. CAGR returned as Null.

Valuation Multiples (P/E & P/B)

Valuation metrics are calculated to assess relative premium or discount:

- P/E Ratio = Stock Price / Earnings Per Share (EPS)
- P/B Ratio = Stock Price / Book Value Per Share (BVPS)

Interest Coverage Ratio (ICR)

ICR assesses how easily a company can pay interest on outstanding debt.

Formula: $ICR = (Profit\ Before\ Tax + Interest\ Expense) / Interest\ Expense$

If Interest Expense is 0, the company is debt-free, and ICR is set to Null.

CFO Quality Score

Measures accrual risk. A score representing average Operating Cash Flow divided by Net Profit over 5 years.

Formula: $Score = Average(CFO) / Average(Net\ Profit)$

Scores > 1.0 indicate *High Quality* cash generation. Scores < 0.5 indicate *Accrual Risk*.

6. REST API Endpoints Reference

The backend is built with FastAPI and runs on port 8000. It exposes 16 endpoints grouped under version /api/v1:

GET /api/v1/health

Health check; returns DB table row counts, server status, and uptime.

GET /api/v1/companies

List all companies; supports query parameters sector, market_cap_category, and search.

GET /api/v1/companies/{ticker}

Get company profile, latest conformed ratios, and AI pros/cons list.

GET /api/v1/companies/{ticker}/pl

Get historical Profit and Loss statements (from_year / to_year).

GET /api/v1/companies/{ticker}/bs

Get historical Balance Sheet statements.

GET /api/v1/companies/{ticker}/cashflow

Get historical Cash Flow statements.

GET /api/v1/companies/{ticker}/ratios

Get historical computed financial ratios.

GET /api/v1/companies/{ticker}/tearsheet

Binary download of pre-generated 2-page ReportLab PDF tearsheet.

GET /api/v1/screener

Apply filters (roe, de, fcf, sector, pe) and get ranked companies by composite quality score.

GET /api/v1/sectors

Summary aggregates for all sectors (company count, median ROE, PE, DE).

GET /api/v1/sectors/{sector}/companies

Get list of companies in a sector with their latest KPIs.

GET /api/v1/peers/{group_name}

Get peer group list with percentile ranks for 10 metrics.

GET /api/v1/companies/{ticker}/peers/compare

Get 8 axes radar comparison data against peer average and benchmark.

GET /api/v1/market-cap/{ticker}

Get historical valuation multiples (2019-2024).

GET /api/v1/portfolio/stats

Get P10 through P90 percentiles, mean, and std for 10 core KPIs.

GET /api/v1/companies/{ticker}/documents

Get annual report links with URL validity syntax flags.

7. Dashboard User Manual

The financial intelligence dashboard is built using Streamlit and provides a premium, responsive visual interface for investment analysts. It has been integrated with the FastAPI REST API, meaning it does not query the database directly. This ensures separation of concerns and lets the REST API handle connection pooling and caching.

Dashboard Structure & Features

The dashboard is structured into four main pages accessible via the sidebar navigation:

- **01 Home Page (Market Overview):** Displays macro indices, top-performing sectors, and a comprehensive table of all 92 companies. Sector summaries (median ROE, PE, D/E) provide high-level context.
- **02 Profile Page (Company Tearsheet):** Focuses on a single selected company. Renders interactive trend line charts (Revenue, PAT, ROE, ROCE) and cash flow waterfall diagrams. Analysts can download the pre-generated 2-page PDF tearsheet via a single download button.
- **03 Screener Page (Stock Filters):** An interactive stock screener. Analysts use sliders and select boxes to filter companies by ROE, leverage, growth, and sector. Matching companies are returned in a table ranked by their *Composite Quality Score*.
- **04 Peers Page (Relative Analysis):** Compares a selected company against its peer group. Renders a radar chart showing the company, peer average, and benchmark leader across 8 operational axes.

Performance & Caching

To ensure sub-second response times, the Streamlit dashboard utilizes `@st.cache_data(ttl=600)` caching on all API client requests. This avoids redundant HTTP traffic and ensures that the dashboard feels fast and responsive during heavy analytical use.

8. Appendix & Data Quality Rules

To guarantee the absolute accuracy of our financial datasets, the platform executes a data quality validation suite consisting of 16 conformed checks. Failures are logged to output/validation_failures.csv.

Rule ID	Description
DQ-01 (CRITICAL)	Company PK Uniqueness: Verifies no duplicate records exist for the same company_id in metadata tables.
DQ-02 (CRITICAL)	Annual PK Uniqueness: Verifies no duplicate rows exist for the same company and fiscal year combination.
DQ-03 (CRITICAL)	Foreign Key Integrity: Verifies all financial statement records reference a valid company_id in the companies table.
DQ-04 (WARNING)	Balance Sheet Balance: Assets and Liabilities must balance within a 1% margin of error.
DQ-05 (WARNING)	OPM Cross-Check: Calculated OPM (Operating Profit / Sales) must match the recorded OPM% within 1%.
DQ-06 (WARNING)	Positive Sales: Non-financial sector companies must report positive sales.
DQ-07 (CRITICAL)	Year Format: Verifies that fiscal year labels conform to the YYYY-MM format.
DQ-08 (CRITICAL)	Ticker Format: Verifies ticker length (2-12 characters) and valid alphanumeric characters.
DQ-09 (WARNING)	Net Cash Check: Cash flow statement CFO + CFI + CFF must equal recorded Net Cash Flow (tolerance ■10 Cr).
DQ-10 (WARNING)	Non-Negative Fixed Assets: Capital assets on the balance sheet must be non-negative.
DQ-11 (WARNING)	Tax Rate Range: Effective corporate tax rate must fall in a reasonable 0% to 60% range.
DQ-12 (WARNING)	Dividend Payout Cap: Dividend payout ratio must not exceed 200%.
DQ-13 (WARNING)	URL Validity: All annual report links must match valid URL formats.
DQ-14 (WARNING)	EPS Sign Consistency: Net profit sign and EPS sign must match.
DQ-15 (INFO)	Strict Balance: Assets and Liabilities must balance to the exact rupee.
DQ-16 (WARNING)	Coverage Check: Every company must have at least 5 years of historical financial records.